VSEC 9 PS — FULL PROCESS (Docker AI Project)

Same commands for both Windows and ubuntu just different environments

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🔹 1. Install Docker

Install Docker

Verify installation

docker --version

👉 Run in:

- Windows → Git Bash
- Ubuntu → Terminal

🔹 2. Create Project Directory

mkdir ai-docker-project
cd ai-docker-project

👉 Run inside terminal

🔹 3. Create Required Files

touch app.py requirements.txt Dockerfile

🔹 4. Write Application Code

Open file

Windows:

notepad app.py

Ubuntu:

nano app.py

Code

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "AI Model Running Successfully!"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

- 5. Create Requirements File

Open:

```
notepad requirements.txt   # Windows
nano requirements.txt      # Ubuntu
```

Add:

```
flask
```

- 6. Create Dockerfile

Open:

```
notepad Dockerfile   # Windows
nano Dockerfile      # Ubuntu
```

Add:

```
FROM python:3.10
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
EXPOSE 5000
CMD ["python", "app.py"]
```

- 7. Build Docker Image

```
docker build -t ai-model-app .
```

👉 Run in project folder

🔹 8. Run Docker Container

docker run -d -p 5000:5000 ai-model-app

🔹 9. Verify Running Container

docker ps

👉 You will see container running

🔹 10. Test Application

Open browser:

http://localhost:5000

👉 Output:

AI Model Running Successfully!

🔹 11. Stop Container

docker ps
docker stop CONTAINER_ID

🔹 12. (Optional) Remove Container

docker rm CONTAINER_ID